MIL-A-9286C(USAF)
27 May 1969
Superseding
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13 May 1957

# MILITARY SPECIFICATION

## AF AMPLIFIER AM-476( )/AIC-10

1. SCOPE.

1.1 This specification covers one type of audio frequency amplifier designated AF Amplifier AM-476( )/AIC-10, one item of Intercommunication Set AN/AIC-10.

1.1.1 In the type designation above, the parenthesis will be replaced by a specific letter, after award of a contract, upon application by the contractor to the procuring activity. The complete nomenclature or type number shall be used on nameplates, shipping records, and in instruction books, as applicable.

2. APPLICABLE DOCUMENTS.

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this document:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-E-5272 | Environmental Testing, Aeronautical and Associated Equipment, General Specification For |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification For |
| MIL-S-19500 | Semiconductor Devices, General Specification For |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements for Equipment |
| MIL-STD-704 | Electric Power, Aircraft, Characteristics and Utilization Of |
| MIL-STD-781 | Reliability Tests Exponential Distribution |

DRAWINGS

US Air Force

| | |
|---|---|
| 58C12600 | Wiring Diagram AM-476C/AIC-10 AF Amplifier, Functional Schematic |
| 53C12734 | Wiring Diagram, AM-476( )/AIC-10, AF Amplifier, Test Circuit |

FSC 5831

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| | |
|---|---|
| 53C12735 | Wiring Diagram, C823A/AIC-10, Interphone Control, AM 476( )/AIC-10 AF Amplifier, Test Circuit, Schematic |
| 53C13654 | AF Amplifier AM-476( )/AIC-10, Outline Dimensions Of |

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS.

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General Specification. The requirements of MIL-E-5400 for class 2 equipment apply as requirements of this document with the exceptions and additions called out herein. When the two documents conflict, this specification shall govern.

3.2.1 Exceptions to the Requirements of MIL-E-5400.

3.2.1.1 Equipment Operation. AF Amplifier AM-476( )/AIC-10 shall operate under the class 2 equipment requirements of MIL-E-5400 except that the maximum temperature for intermittent and continuous operation shall be +71°C. The temperature vs altitude curves of the operational requirement for class 2 airborne electronic equipment shall be adjusted accordingly.

3.2.1.2 Substitution of Material. If a manufacturer, either prior to or during production, desires to substitute another material or process for one previously approved or specified, he shall submit to the procuring activity a statement to that effect. The statement shall describe the proposed substitution and the reason therefor, together with evidence to show that the substitution is satisfactory. At the discretion of the procuring activity, test samples or additional data may be required to prove the adequacy of the proposed substitution.

3.2.1.3 Transistors. The transistors shall meet the requirements of MIL-S-19500.

3.2.1.4 Construction. The construction shall be such that the amplifier is inherently stable as to mechanical construction and electrical performance. Encapsulation or hermetic sealing shall be employed, even though this will preclude repair.

3.2.1.5 Encapsulating Material. Cured samples of the encapsulating material employed shall show no evidence of cracking after 24 hours of exposure at -65°C and no evidence of cracking or fusion after 24 hours of exposure at +105°C. Materials having moisture absorption exceeding 1.04 percent shall be impregnated, after curing, with a suitable varnish. In addition, the properties of the encapsulating material shall be such that the encapsulated amplifier shall show no cracks or other evidence of deterioration when the completed amplifier assembly is subjected to the environmental conditions and



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temperature shock tests specified herein.

3.2.1.6 Service Conditions (Electrical). The equipment shall be designed to operate from a power source with electrical characteristics conforming to the dc power requirements of MIL-STD-704 Category B, Table II steady state limits.

3.3 Electrical Circuit. The functional circuit schematic of the AM-476( )/AIC-10, as shown in Drawing 58C12600, shall be used as a reference to determine the general circuitry and components of the amplifier. The contractor has the option to deviate from the circuit given, provided the functional characteristics of this specification are met.

3.4 Interference. Electromagnetic interference control shall be in accordance with applicable requirements of MIL-STD-461, Table I, for Class IC equipment.

3.5 Reliability. The contract specified mean-time-between-failure (MTBF) shall not be less than 7500 hours.

3.6 Performance.

3.6.1 Performance Requirements as a Headset Amplifier. When operating as a headset amplifier, the amplifier shall provide the following performance when tested as specified herein.

3.6.1.1 Input-Output Characteristic. The input-output characteristic at 1000 Hz shall be in accordance with table I.

TABLE I

| INPUT VOLTS TO 150-OHM PAD | NOMINAL OUTPUT VOLTS | MINIMUM OUTPUT VOLTS | MAXIMUM OUTPUT VOLTS |
|---|---|---|---|
| 0.012 | 2.25 | 1.35 | 3.50 |
| 0.017 | 3.10 | 1.90 | 4.90 |
| 0.021 | 3.75 | 2.35 | 6.75 |
| 0.038 | 6.40 | 4.40 | 8.50 |
| 0.120 | 7.40 | 5.80 | 9.00 |
| 0.214 | 7.50 | 5.90 | 9.20 |

3.6.1.2 Frequency Response. With a constant input of 0.012 volt to the 150-ohm input pad specified herein, the output of the amplifier, relative to the 1000 Hz output, shall be within the limits specified in table II.



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TABLE II

| FREQUENCY (Hz) | OUTPUT (DB RELATIVE TO 1000 Hz) |
|---|---|
| 300 | plus 0, minus 4.5 |
| 500 | plus 0, minus 3.5 |
| 1000 | 0 |
| 1500 | plus 1.0, minus 1.0 |
| 2000 | plus 1.0, minus 1.0 |
| 3000 | plus 1.0, minus 1.0 |
| 4000 | plus 1.0, minus 3.0 |
| 5000 | plus 1.0, minus 4.5 |
| 6000 | plus 0, minus 6.0 |
| 10000 | minus 5.0, minus 15.0 |

3.6.1.3 Automatic Gain Control (AGC) Disabling. The AGC circuit shall be inoperative and shall not generate and store a control voltage during operation of the amplifier as a headset amplifier.

3.6.1.4 Power Output. The maximum power output of the amplifier at 1000 Hz shall be at least 0.280 watt.

3.6.1.5 Distortion. The total harmonic distortion present in the output voltage of the amplifier at 300 Hz shall not exceed the values specified in table III.

TABLE III

| OUTPUT VOLTS | MAXIMUM DISTORTION (%) |
|---|---|
| 1.37 | 4 |
| 1.73 | 4 |
| 2.18 | 4 |
| 2.74 | 4 |
| 3.2 | 4 |
| 4.0 | 6 |
| 5.0 | 12 |

3.6.1.6 Microphonic Noise. When subjected to standard vibration cycling, the microphonic noise output of the amplifier shall not exceed 0.018 volt.

3.6.1.7 Phasing. Circuit connections and transformer windings of the amplifier shall be such that the output signals are 180 degrees ±15 degrees out of phase with the input signals.

3.6.2 Performance Requirements as a Microphone Amplifier. When operating as a microphone amplifier, the amplifier shall provide the following performance when tested as specified herein.

3.6.2.1 Input-Output Characteristic. The input-output characteristic at 1000 Hz shall be in accordance with table IV.



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TABLE IV

| INPUT VOLTS TO 5-OHM PAD | NOMINAL OUTPUT VOLTS | MINIMUM OUTPUT VOLTS | MAXIMUM OUTPUT VOLTS |
|---|---|---|---|
| 0.024 | 0.16 | 0.09 | 0.29 |
| 0.0425 | 0.29 | 0.16 | 0.52 |
| 0.075 | 0.52 | 0.29 | 0.92 |
| 0.134 | 0.92 | 0.52 | 1.63 |
| 0.24 | 1.46 | 0.92 | 2.06 |
| 0.425 | 1.63 | 1.15 | 2.31 |
| 0.75 | 1.83 | 1.15 | 2.44 |
| 1.34 | 1.94 | 1.15 | 2.59 |
| 2.4 | 2.06 | 1.15 | 2.91 |
| 4.25 | 2.31 | 1.15 | 3.08 |

3.6.2.2 Frequency Response. With a constant input of 0.07 volt to the 5-ohm pad specified herein, the output of the amplifier relative to the 1000 Hz output shall be within the limits specified in table II.

3.6.2.3 Distortion. The total harmonic distortion present in the output voltage of the amplifier at 300 Hz shall not exceed the values shown in table V.

TABLE V

| INPUT VOLTS TO 5-OHM PAD | MAXIMUM DISTORTION IN % |
|---|---|
| 0.134 | 3 |
| 0.425 | 3 |
| 0.75 | 5 |
| 1.34 | 9 |
| 2.4 | 15 |

3.6.2.4 Signal-to-Noise Ratio. The signal-to-noise ratio of the amplifier shall be not less than 50 db when tested in accordance with paragraph 4.5.2.2.4.

3.6.2.5 Automatic Gain Control (AGC). The release time of the AGC circuit shall be 7 seconds plus or minus 2.5 seconds. The AGC attack time shall not exceed 0.20 second.

3.6.2.5.1 Location of AGC Breakpoint. The AGC breakpoint is defined herein as the knee of the input-output curve caused by the AGC action of the amplifier. The breakpoint shall be the point where a variance in linearity of the input-output characteristic is 1 db. This point shall occur at an input voltage between 0.170 and 0.30 volt inclusive.

3.6.2.5.2 AGC Slope. The AGC slope is defined herein as the slope in the input-output curve in the region beyond the AGC breakpoint. The slope shall be such that the increase in the output of the amplifier shall be not more than 4 db when the input to the 5-ohm input pad specified herein is increased from 0.45 volt to 1.41 volts.



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3.6.2.5.3 Output-to-Input Feedback. Circuit connections and transformer windings of the amplifier shall be such that a capacitance of 0.001 mfd connected between either side of the output and the 5-ohm input terminals shall not cause the amplifier to oscillate. Terminate the output of the amplifier into a 1000 ohm load and monitor the output wave with an appropriate oscilloscope. There shall be no evidence of any oscillation.

3.6.3 Performance Requirement as a Special Amplifier. With an input of 0.1 volt to the 2.5-ohm input pad specified herein, the output of the amplifier at 1000 Hz shall be 0.3 volt plus 5 db, minus 4 db.

3.6.4 High-Voltage Load Simulator. The resistive loads connected between terminals "E" and "D" of J102 as shown on Drawing 58C12600 shall satisfactorily dissipate 4 watts of power during exposure to any environmental conditions without damage to the load or to the amplifier.

3.7 Weight. The weight of the amplifier shall not exceed 0.95 pound.

3.8 Dimensions. The dimensions of the amplifier shall be in accordance with Drawing No. 53C13654.

3.9 Compatibility. The amplifier shall be electrically and physically compatible with Controls C-823, C-824, C-825/AIC-10, and Relay Assembly RE94A/AIC-10.

3.10 Government-Loaned Property. When provided for in the contract or purchase order, and if available, the following items will be furnished by the Government on loan to the contractor, in the quantities required, upon his request.

| Item Number | Description |
|---|---|
| 1 | Control Panel C-824A/AIC-10<br>(Includes AF Amplifier AM-476( )/AIC-10) |
| 2 | Headset-Microphone H-78( )/AIC |
| 3 | Interphone Control C-823A/AIC-10<br>(Includes AF Amplifier AM-476( )/AIC-10) |
| 4 | Interphone Control C-825/AIC-10 |
| 5 | Relay Assembly, RE94A/AIC-10<br>(Includes AF Amplifier AM-476( )/AIC-10) |

4. QUALITY ASSURANCE PROVISIONS.

4.1 Classification of Tests. The inspection and testing of AM Amplifier AM-476( )/AIC-10 shall be classified as follows:

a. Preproduction tests . . . . . See 4.3

b. Acceptance tests . . . . . See 4.4

4.2 Test Conditions. Unless otherwise specified, the amplifiers shall be tested under the following conditions:



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a. Temperature: Room ambient, plus 15°C to plus 35°C

b. Pressure: Normal atmospheric

c. Humidity: Room ambient, up to 90 percent relative humidity

d. Primary voltage supply: 27.5 volts dc ± 1.0 volt.

4.3 Preproduction Tests. Preproduction tests shall be performed by the contractor in accordance with the following:

4.3.1 Preproduction Test Samples.

4.3.1.1 Contractor Test Samples. When specified in the contract a minimum of 25 each AF Amplifiers AM-476( )/AIC-10 shall be subjected by the contractor to the preproduction tests specified herein. Twenty-two (22) of these samples shall be used for Reliability Testing only.

4.3.1.2 Government Preproduction Test Samples. When specified in the contract, the 3 each amplifiers not used above in reliability testing shall be submitted to the procuring activity. The sample amplifiers shall be accompanied by adequate identifying information including the Government order number involved ("File Number" in contract), if any, name and address of prime contractor, and name and address of manufacturer. The samples and data will be used by the procuring activity for:

a. Review of the mechanical design.

b. Performance of any tests, specified herein, which may be deemed necessary after a review of the contractor's test report.

c. Government reference purposes.

4.3.2 Preproduction Test Routine. Preproduction samples shall be subject to the tests listed in table VI in the quantities shown.

TABLE VI

| INSPECTION OR TEST | TEST PROCEDURE OR REQUIREMENT | PREPRODUCTION | INDIVIDUAL | SAMPLING |
|---|---|---|---|---|
| Mechanical Inspection | 4.5.1 | 3 | all | all |
| Electrical Tests | 4.5.2 | | | |
| Headset Amplifier | 3.6.1 | | | |
| Input-Output Characteristic | 3.6.1.1 | 3 | | all |
| Input-Output Characteristic (Acceptance Tests) | 4.5.2.1.2 | | all | |
| Frequency Response | 3.6.1.2 | 3 | | all |
| Frequency Response (Acceptance Tests) | 4.5.2.1.3 | | all | |
| Power Output | 3.6.1.4 | 3 | | all |
| Distortion | 3.6.1.5 | 3 | | all |



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TABLE VI (Cont'd)

| INSPECTION OR TEST | TEST PROCEDURE OR REQUIREMENT | PREPRODUCTION | INDIVIDUAL | SAMPLING |
|---|---|---|---|---|
| Distortion (Acceptance Test) | 4.5.2.1.4 | | all | |
| Microphonic Noise | 4.5.2.1.5 | 3 | | all |
| Phasing | 4.5.2.1.6 | 3 | | all |
| Microphone Amplifier | 3.6.2 | | | |
| Input-Output | 3.6.2.1 | 3 | | all |
| Input-Output (Acceptance Test) | 4.5.2.2.2 | | all | |
| Frequency Response | 3.6.2.2 | 3 | | all |
| Frequency Response (Acceptance Test) | 4.5.2.2.3 | | all | |
| Distortion | 3.6.2.3 | 3 | | all |
| Signal-To-Noise Ratio | 3.6.2.4 | 3 | all | all |
| AGC Attack Time | 4.5.2.2.5 | 3 | | all |
| AGC Release Time | 4.5.2.2.6 | 3 | | all |
| AGC Slope | 4.5.2.2.7 | 3 | all | all |
| Output-To-Input Feedback | 3.6.2.5.3 | 3 | | all |
| Special Microphone Amplifier | 4.5.2.3 | 3 | | all |
| Performance Requirement as a Special Amplifier | 3.6.3 | 3 | | all |
| Talk-Listen Tests | 4.5.2.4 | 3 | all | all |
| Interference | 3.4 | 3 | | |
| Load-Simulator Continuity | 3.6.4 | | | |
| Load Simulator Continuity Test | 4.5.2.6 | 3 | all | all |
| Reliability | 3.5 | | | |
| Reliability Tests | 4.5.2.5 | 22 | | |
| Environmental Tests | 4.5.3 | | | |
| Fungus | | 2 | | |
| Shock | | 2 | | |
| Vibration | | 2 | | |
| Humidity | | 2 | | |
| Salt Spray | | 2 | | |
| Temperature Altitude | | 2 | | all |
| Temperature Shock | | 2 | | all |

NOTE: Where a paragraph in Section 3 is referenced above, the test procedure shall consist of determining complete compliance with the requirements stated in that paragraph.

4.4 Acceptance Tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests



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4.4.1 Individual Tests. Each amplifier shall be subjected to the individual tests listed in table VI.

4.4.2 Sampling Tests. Unless otherwise specified, sampling tests shall be conducted on at least one out of every 100 amplifiers on order. The samples shall be selected at random from quantities of amplifiers which have passed the individual tests. The samples shall be subjected to the sampling tests listed in table VI in the quantities shown.

4.4.2.1 Acceptance Criteria. When one or more of the selected amplifiers fails to meet the specified requirements, acceptance of all items shall be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5 Test Procedures.

4.5.1 Examination of Product. The amplifier shall be examined to determine compliance with the requirements of this specification for materials, workmanship, dimensions, and markings using whatever tools, gauges, scales, and the like, that are necessary.

4.5.2 Electrical Tests. The amplifier shall be electrically tested in order to determine compliance with the performance requirements of this specification. The test circuits and pads used shall be in accordance with Drawing No. 53C12734.

4.5.2.1 Headset Amplifier.

4.5.2.1.1 Headset Amplifier Test Circuit Conditions. The test circuit conditions shall be as follows:

a. The 150-ohm input pad shall be connected to the 150-ohm input circuit of the amplifier.

b. The AGC terminal of the amplifier shall be connected to ground.

c. The gain reduction terminal shall be connected to ground through a 1000-ohm resistor.

d. The amplifier load shall be 150 ohms.

e. Input voltage measurements shall be made at the input of the 150-ohm pad.

f. Output voltage measurements shall be made across the 150-ohm load resistor.

4.5.2.1.2 Headset Amplifier Input-Output Characteristic. The amplifier output shall be measured at 1000 Hz with an input of 0.012 volt to the 150-ohm pad.



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4.5.2.1.3 Headset Amplifier Frequency Response. The response shall be at 300 and 6000 Hz with an input of 0.012 volt to the 150-ohm pad.

4.5.2.1.4 Distortion. The distortion in the output of the amplifier shall be measured when the amplifier is developing 4.0 volts across the load resistor at 300 Hz.

4.5.2.1.5 Microphonic Noise Test. This test shall be performed with the amplifier mounted in an Interphone Control C-823A/AIC-10. The control shall be wired in accordance with Drawing 53C12735, with a 150-ohm resistor connected across terminals K and J, and a Headset-Microphone H-78( )/AIC connected to the C-823A/AIC-10 as shown on the drawing. The C-823A/AIC-10 shall then be vibrated through three complete vibration cycles of 10 to 55 Hz at a double amplitude of 0.06 inch at all frequencies. The vibration excursion shall be in the direction of the toggle switch arrows on the C-823A/AIC-10. The press-to-talk switch shall be open during vibration. The maximum voltage reading across the headset during the third complete vibration cycle shall be taken as the microphonic noise output of the amplifier.

4.5.2.1.6 Headset Amplifier Phasing. The phasing shall be checked to insure that the instantaneous voltage between terminal F and terminal A of J-101, is 180 degrees ± 15 degrees out of phase with the instantaneous voltage between terminal H and terminal C of J-102.

4.5.2.2 Microphone Amplifier.

4.5.2.2.1 Microphone Amplifier Test Circuit Conditions. The test circuit conditions shall be as follows:

a. The 5-ohm pad shall be connected to the 5-ohm input circuit of the amplifier.

b. The AGC terminal shall be open.

c. The gain reduction terminal shall be connected to ground.

d. The amplifier load shall be 75 ohms.

e. Input voltage measurements shall be made at the input of the 5-ohm pad.

f. Output voltage measurements shall be made across the 75-ohm load.

4.5.2.2.2 Microphone Amplifier Input-Output Characteristic. The amplifier output shall be measured at 1000 Hz with an input of 0.075 volt to the 5-ohm pad.

4.5.2.2.3 Microphone Amplifier Frequency Response. The response shall be measured at 300 and 6000 Hz with an input of 0.075 volt to the 5-ohm pad.



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4.5.2.2.4 Signal-to-Noise Ratio. The normal output shall be determined by measuring the output voltage across the 75-ohm load resistor, with a 0.075 volt signal applied to the 5-ohm input pad and while the noise conditions specified below are also present. The noise output shall then be measured under each of the following conditions:

a. With no signal applied to the input and with a pure dc power source.

b. With no input signal and with 1.5 volts (rms) at 400 Hz superimposed on the 27.5 volt dc supply line.

4.5.2.2.5 AGC Attack Time. The AGC attack time shall be measured by suddenly applying a 1000 Hz input of 1.3 volts (rms) to the 5-ohm input pad. The time required for the output voltage to rise to within 2 db of its final value is the AGC attack time.

4.5.2.2.6 AGC Release Time. The AGC release time shall be measured by applying a 1000 Hz input of 1.3 volts (rms) to the 5-ohm pad for at least 2 seconds and then instantly reducing the input by 10 db. The AGC release time is the time required for the output to return to within 4 db of its final value.

4.5.2.2.7 AGC Slope. The AGC slope shall be determined by applying a 1000 Hz input of 0.45 volt to the 5-ohm pad and then increasing the input to 1.41 volts.

4.5.2.3 Special Microphone Amplifier Test Circuit Conditions. The test circuit conditions shall be as follows:

a. The 2.5-ohm pad shall be connected to the 5-ohm input circuit of the amplifier.

b. The AGC terminal shall be connected to ground.

c. The gain reduction terminal shall be connected to ground.

d. A 610-ohm resistor shall be connected in series between terminal H of J-102 connector and a 150-ohm load resistor.

e. Input voltage measurements shall be made at the input to the 2.5-ohm pad.

f. Output voltage measurements shall be made across the 150-ohm load resistor.

4.5.2.4 Talk-Listen Test. Two amplifiers shall be installed in C-823A/AIC-10 Interphone Controls connected in accordance with Drawing No. 53C12735. Each amplifier shall be tested for satisfactory operation both as a headset amplifier and as a microphone amplifier.

4.5.2.5 Reliability Preproduction Tests.



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4.5.2.5.1 Reliability Test. Reliability tests as required by paragraph 4.3.2 shall be performed to Test Level F from MIL-STD-781.

4.5.2.5.2 Temperature Cycling. Temperature cycling shall be Standard Method of Temperature Cycling as specified in MIL-STD-781. The amplifier shall be maintained at the upper test level temperature for a period of three (3) hours.

4.5.2.5.3 Specific Duty Cycle. Specific duty cycle shall be in accordance with MIL-STD-781, with the amplifier operating during alternate "ON" cycles as: (a) microphone amplifier (b) headset amplifier.

4.5.2.5.4 Voltage Cycling. Voltage cycling shall be in accordance with MIL-STD-781.

4.5.2.5.5 Equipment Operating Parameters. The operating parameters shall be as specified in the individual tests of table VI. Failure to meet the minimum requirements specified shall constitute a failure. Measurement of operating parameter shall be in accordance with MIL-STD-781.

4.5.2.5.6 Test Plan. Test Plan V is to be performed in accordance with MIL-STD-781, Table 2 with $\theta_0 = 7500$, $\theta_1 = 2500$.

4.5.2.5.7 Test Option. The contractor has the option of running the reliability preproduction tests simultaneously with the other required preproduction tests with the stipulation that if any change is found needed by the other tests the reliability test must be restarted with zero time. If a failure occurs it will be analyzed, repaired and replaced on test.

4.5.2.6 Load-Simulator Continuity Test. With no voltage applied to the amplifier, an ohmmeter shall be connected to terminals "E" and "D" of Plug P3002, of test circuit shown on Drawing 53C12734, to measure the resistance of the high voltage load simulator. Resistance shall be with the resistance tolerances of the load resistors used.

4.5.3 Environmental Tests. The environmental tests listed in table VII shall be in accordance with MIL-E-5272, and the instructions indicated below. During shock, vibration, and temperature tests, the amplifier shall be mounted in an Interphone Control C-823A/AIC-10. The rejection criteria for the unit shall be any deterioration of performance which would prevent the amplifier from meeting the minimum requirements of the individual tests of table VI.

TABLE VII

| TEST | PROCEDURE |
|---|---|
| Fungus | I |
| Shock | V |
| Vibration | XII |
| Humidity | I |
| Salt Spray | I |
| Temperature Altitude | I |
| Temperature Shock | I |



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5. PREPARATION FOR DELIVERY.

5.1 General. Preparation for delivery shall be in accordance with provision specified in procurement contract.

6. NOTES.

6.1 Use. Amplifier AM-476( )/AIC-10 is an audio frequency amplifier. It is a component of Interphone Control C-823A/AIC-10, Control Panel C-824A/AIC-10, Control Panel C-825A/AIC-10, and Relay Assembly RE-94A/AIC-10, all of which are items of Intercommunication Set AN/AIC-10. The AM-476( )/AIC-10 amplifier is intended for use in military aircraft both as a microphone preamplifier and a headset amplifier.

6.2 Ordering Data. Invitations for bids, contracts, and purchase order will specify that 25 each AF Amplifier AM-476( )/AIC-10 shall be required as preproduction samples to be subjected to the preproduction tests to determine compliance with the requirements of this specification, and that the 3 amplifiers AM-476( )/AIC-10 not used in reliability testing shall be submitted to the procuring activity. The invitation for bids and the contract should specify the point of inspection for these tests.

6.3 Changes From Previous Issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extentiveness of the changes.

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Air Force - 84

Preparing Activity:
Air Force - 84

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE C-0070

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